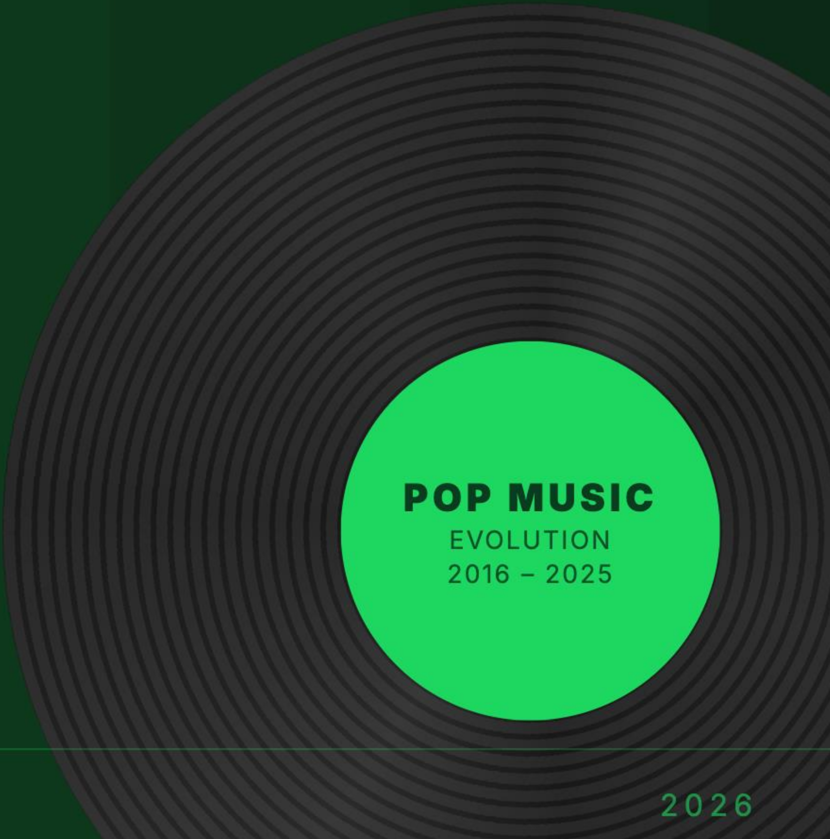
THE SOUND OF A DECADE

POP MUSIC EVOLUTION

Songs are shorter. Genres are blending.

Explore how music changed.



EPFL DATA VISUALIZATION

2026

MusicEnjoyers

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0. Our Path

In Milestone 1, we each came up with some topic ideas and discussed them. In the end, we found Wei-En's proposal "exploring how popular music has changed over time" the most interesting, and chose it as our topic.

We then used Python to query data from Spotify and built our own dataset of the most popular songs over the past decade. At the same time, we surveyed related work and ran a series of preliminary data-analysis experiments in Python. From these, we gathered our early insights and a clearer sense of what was worth visualizing.

Because our dataset was high-dimensional (covering everything from duration and tempo to danceability, energy, and genre) we explored many different possibilities for what we could ultimately present. We settled on the directions described in this report: simpler song structure, music genre groupings, musical-feature analysis, and lyrical complexity.

In Milestone 2, we realized that we could not present too many themes at once, so we decided to refocus and tighten our storyline. Wei-En's original motivation for the topic was to examine how, following the rise of short-form video, popular music has gradually become shorter, more similar to one another, and less distinctive. With this in mind, we designed our website around three stages: first exploring song duration, then genre, and finally an interactive dashboard where users could engage with the data themselves.

Before building, we sketched out the design of each section by hand (as shown in the Milestone 2 report). This made it much easier to discuss our ideas and gave everyone a shared vision of how the site would look and flow.

We also began implementing the website during this milestone. We built a basic prototype using HTML and JavaScript and started generating our first charts with D3. To ensure the site could render our data quickly, we used Python to clean and process our database in advance and stored the result as static JSON files on the server, which let us skip back-end preprocessing at runtime.

In Milestone 3, we implemented all of the visualizations we had proposed in Milestone 2. We then made a number of refinements to the overall design of the site (repositioning the header, redesigning the cover, and adjusting the placement of the UI elements) before arriving at our final result. That marked the end of our process: from initial ideas, to a clear storyline, to the final website.

1. Introduction & Research Question

"Songs are shorter. Genres are blending. Explore how music changed."

Since TikTok launched in September 2016, the average Billboard Hot 100 has lost more than 10 seconds. Instagram Reels and YouTube Shorts followed, rewarding brevity and punishing anything that takes more than three seconds to land its hook. The attention economy did not just change how we listen to pop music: it reshaped how pop music is written.

But shorter songs are only the most visible symptom. As streaming flattened the gatekeepers that used to separate scenes, genres started bleeding into each other. So we asked:

Has a decade of streaming made popular music sound more alike, or has it actually opened the charts to a wider range of sounds?

To answer this, we analyzed Billboard Hot 100 tracks from 2016 to 2025, enriched with Spotify audio features (tempo, energy, danceability, valence, acousticness, and more). The visualization is structured around three sections: Duration, Genres, and Dashboard.

2. Data & Process

To ask whether pop music is converging or diversifying, we needed two things the charts alone can't tell us: what was popular, and what it actually sounded like.

So we joined two sources. The Billboard Hot 100 year-end charts from 2016 to 2025 told us which songs broke through. Spotify's audio features (tempo, energy, danceability, valence, and a dozen others) told us how those songs were built. Genre tags came along with the Spotify side, which is what later let us ask whether the sound of pop is converging even as the labels multiply.

Joining them was harder than expected. Billboard lists tracks as written ("Bad and Boujee (feat. Lil Uzi Vert)"); Spotify stores them under a different title, encoding, or featured-artist convention. We rely on fuzzy string matching to bridge the two, and we log the unmatched cases so we can spot-check what slipped through.

The site itself runs on static, pre-aggregated JSON files, one per chart. Heavy lifting (deduplication, clustering, feature-rescaling for the radar, the K-Means used in the Similarity view) happens offline; the browser only renders. That choice keeps the page fast and lets the whole thing live on a static host, no server-side computation required at request time.

3. Design Decisions

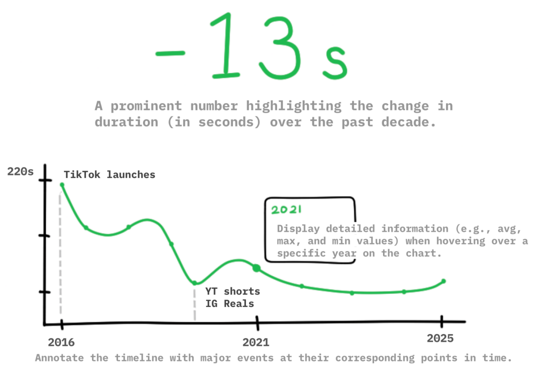
3.0 The main theme

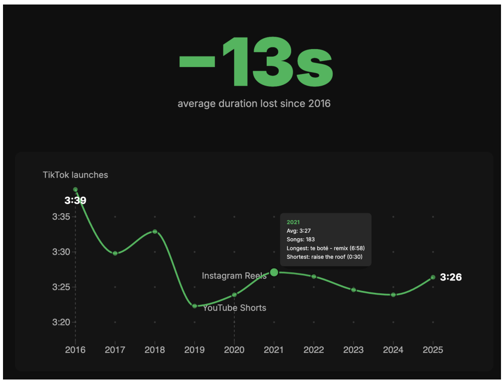
We took inspiration from Spotify's visual identity — a dark background with bright green accents — to immediately signal that this is a music-focused experience and to give the site a modern, polished feel. The cover page opens with floating musical notes and the headline "POP MUSIC IS EVOLVING," evoking a sense of history unfolding — a reminder that what we hear today is the result of a decade of shifting tastes, cultural moments, and evolving trends.
The website is then structured into three clear sections, each answering one question: 1) Are songs getting shorter? 2) How are genres shifting? 3) What does the data say? This progression gives the reader a natural journey through the story.

3.1 Duration: from simple line chart to annotated timeline

We open with a single bold number, **−13 seconds**, the average duration lost since 2016. This anchors the narrative before the chart appears.
The line chart then shows average song duration per year, with 2016 (3:39) and 2025 (3:26) explicitly labeled as endpoints. Three key events are annotated on the timeline: TikTok (September 2016), Instagram Reels (2020), and YouTube Shorts (2020), so users can draw their own conclusions about the trend.

Hovering on any year's dot reveals a tooltip with: average duration, total number of songs, and the longest and shortest songs with their lengths.



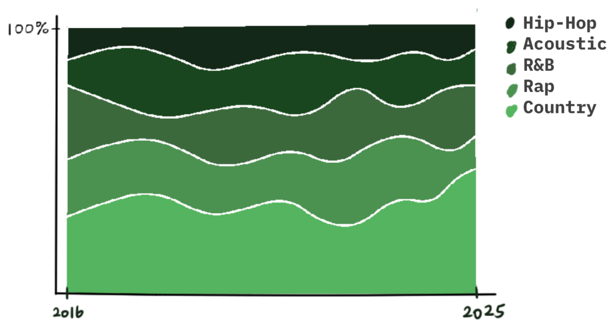


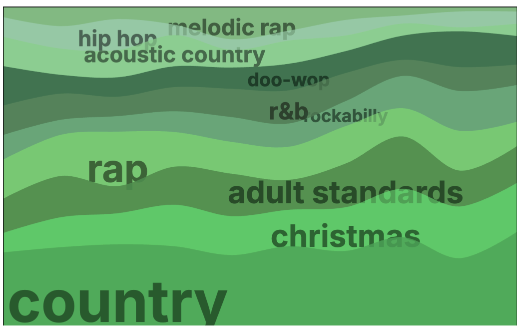
3.2 Shifting Genres and Popularity Trends

In this section, we explore how pop music genres and their audio features evolve over time. Each song in our dataset carries both a genre label (e.g. country, rap, hip-hop) and a set of audio feature values (e.g. Energy, Danceability, Tempo).

We use four visualizations to tell this story, moving from genre-level population trends down to the audio feature profile of the songs themselves.

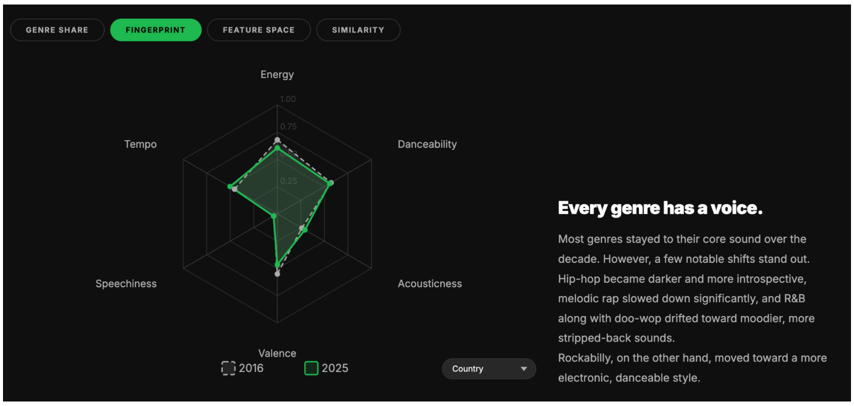
1. **Genre Share** uses a streamgraph to show how each genre's share of the Billboard Hot 100 changes across 2016–2025. Compared to our Milestone 2 design, we moved the labels directly onto the chart instead of using a separate legend, making it much easier to identify each genre at a glance.



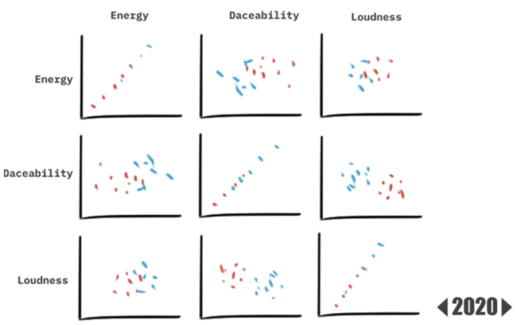


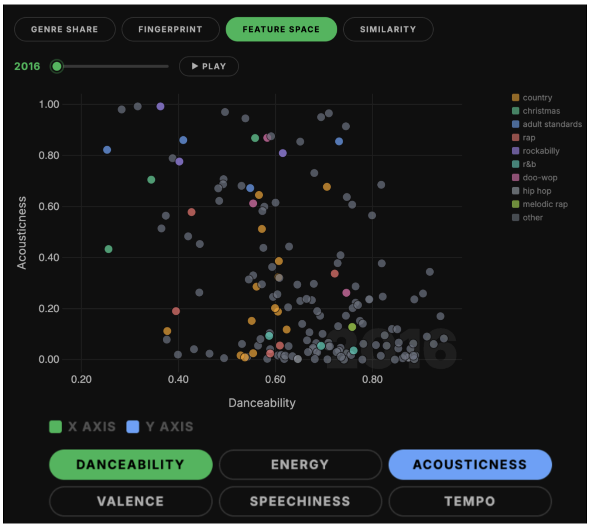
2. **Fingerprint** is a new chart added since Milestone 2. We wanted to bridge genres and audio features, so we used a radar chart to

compare each genre's audio profile in 2016 vs. 2025 across six dimensions — Energy, Danceability, Tempo, Speechiness, Acousticness, and Valence. Users can select any genre from a dropdown to see the fingerprint of that genre and how it changed over the decade.

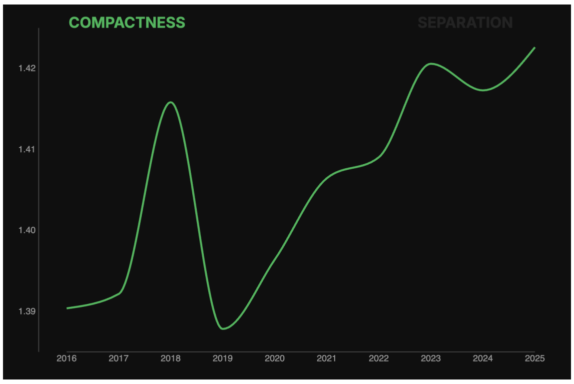


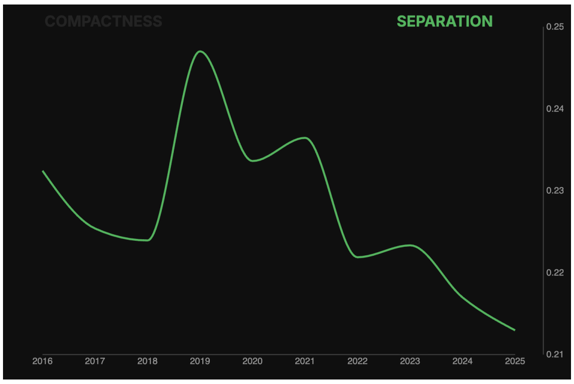
3. **Feature Space** plots every song as a dot in a 2D space where the axes are audio features. Users can pick which features go on each axis and drag a year slider to see how the overall distribution shifts over time. In Milestone 2 we planned multiple small sub-charts, but this made each chart too small to read, so we switched to a single interactive view instead.



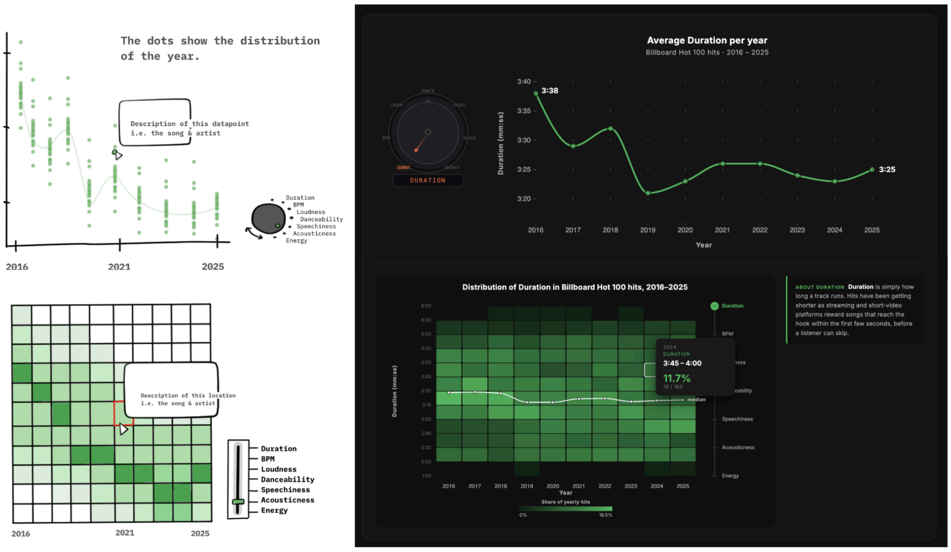


4. **Similarity** clusters songs in each year via k-means (k=5) across Danceability, Energy, Tempo, Valence, and Acousticness. Compactness (avg. distance to centroid) and Separation (silhouette score) both decline over time — songs within clusters grow more alike, while clusters themselves become harder to tell apart. Together, these trends reveal that pop music's sonic palette is converging: genres are blending into a shared sound.





3.3 Dashboard: from dropdown to knob



The demo website in Milestone 2 used a standard dropdown to switch between audio features on the dashboard chart. We replaced it with a rotary knob we designed in milestone 2: same function, but the interaction echoes the subject matter. It is a small flourish, but it also gives the dashboard a strong visual anchor that the dropdown never had, and it makes the section feel like a control panel rather than a form.

In addition, we also reformat the dashboard, so without scrolling, we can visualize the line chart and the heatmap data based on the audio feature we select.

4. Challenges

4.1 Matching Billboard to Spotify

We query data of Billboard Hot 100 tracks from 2016 to 2025 on Billboard, but they do not contain all the audio features we want as on Spotify, thus we had to query each song on Spotify to find its audio features such as tempo, energy, danceability, valence, acousticness, etc.

4.2 Setting our storyline

Our dataset covers a dozen audio dimensions across ten years of chart hits. If we tried to show everything, nothing landed. Every chart felt like it needed a paragraph of explanation just to make sense. So we decide to focus on our central questions first: has a decade of streaming made popular music sound more alike, or has it actually opened the charts to a wider range of sounds?

So we adopted a three-stage strategy: the first two parts tell the story we wanted to share. The final part gives the Dashboard for users to explore freely, where all the data is available without overwhelming the main story.

4.3 Making abstract audio features feel real

Numbers like Danceability, Valence, and Speechiness are not meaningful to most users. They are just labels on a scale. Showing a raw value of 0.73 means nothing. Our solution was the Fingerprint radar chart. Instead of presenting the numbers directly, we turned each genre's audio profile into a shape. Users can combine these numbers with music genres (country music, hip-hop, rap) they are familiar with.

4.4 Keeping everything visually coherent

We ended up with a lot of different chart types. To hold it all together, we committed early to a single visual language inspired by Spotify: dark backgrounds with bright green as the primary accent. Every chart uses the same color palette and typography, so no matter which section is in view, it still feels like one cohesive product rather than a collection of separate visualizations.

5. Peer assessment

Li An-Jie

I helped with the Exploratory Data Analysis in Milestone 1 and collected related works. In Milestones 2 and 3, I was mainly responsible for the second part of our three-stage structure (Shifting Genres and Popularity Trends), created the sketches, and compiled the report. On the implementation side, I set up the demo website in Milestone 2 and completed the design and implementation of the Part 2 visualizations and the cover page in Milestone 3. I was really happy to work with Wei-En and Kenji on such an interesting topic.

Rohner Kenji

The biggest part of my contribution was building the scraper to create our dataset. Aside from our small initial dataset, we needed to link the top 100 Billboard songs to the data/API of Spotify. I was responsible for the scraping, formatting, and matching of two different datasets.

In addition to contributing to the brainstorming, I was also responsible for the last visualization, the line graph, and the heatmap, together with the buttons used for it. I also wrote another script that took Wei-En’s script as a base to preprocess the data and make it available as a JSON file.

Hsieh Wei-En

Milestone 1: I proposed the project's research question. I built the first version of the working dataset from the Billboard Hot 100, producing a CSV using the Deezer API for the album name, duration, and explicit fields, then manually resolved the entries the automation could not match. I also produced the early per-year duration analysis that became the project's headline finding.

Milestones 2 and 3: I restructured the project from a three-page site into a single scrolling page with D3 section transitions. I designed and implemented Part 1 (Duration) end to end. I also partnered with Kenji for part 3: restructuring the section's layout, aligning the Part 3 line chart with Part 1's, refining the knob interaction, writing the per-feature explanations, rebuilding the heatmap as a percent-of-yearly-total view with a gradient legend, a weighted median trend line, and tooltips showing each value's unit and its delta against a baseline year. It was a pleasure working with An-Jie and Kenji on a topic I care about.